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SOME PALEOMAGNETIC RESULTS FROM RAROTONGA, COOK ISLANDS

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ABSTRACT

Lavas and dikes in Rarotonga were sampled at 12 sites and are all found to be of reversed polarity. The mean directions of remanence are found to be 223°, +59°, for samples from the Avatiu Valley and 180°, +48°, for samples from the Tuoro quarry; the average circular standard deviations are 47° and 20° respectively. Insufficient sampling prevents interpretation in terms of secular variation.

INTRODUCTION

Rarotonga is the largest (250 km²) of the islands forming the Cook Islands (Fig. 1). The island is rugged, numerous peaks rising over 500 m, but the highest point is only 650 m. The volcanic hills are surrounded by a narrow coastal strip of coral sand and other beach deposits, usually separated from the hills by an elongated swampy depression. The coastal strip, which consists in part of raised reef and beach material, supports most of the island's population and provides the greatest area of flat land.

Marshall (1930) described the volcanic geology in some detail, but presented only a small locality map. He inferred an alternating sequence of basaltic and phonolitic eruptions. More recent work by Wood and Hay (1967), illustrated by a geological map, shows that the strongly dissected landforms still display the elements of the major volcanic structures—a large basaltic cone, with younger phonolitic eruptives occupying a caldera. The latest eruptions are considered to be probably as young as late Tertiary or early Pleistocene, like those of strongly dissected Oahu, Hawaiian Islands (McDougall, 1964; McDougall and Tarling, 1963). Dr N. H. Gale of Oxford University has (pers. comm.) made age estimates based on whole-rock potassium/argon analyses of sample 10b from site 9 and sample 22 from site 12b; these are 2·30 ±0·24, and 2·80 ±0·13 million years respectively and correspond to a late Pliocene age, thereby supporting Marshall's comparison with Oahu.

As a part of a reconnaissance study of the paleomagnetism of Late Cenozoic basalts in and around the Pacific (Tarling, 1963, 1965, 1966a and b, 1967) the Cook Islands were visited briefly in 1960. The exposures and time available were both limited and only 12 sites (Fig. 1) in lavas and dikes were sampled, mostly in the Avatiu Valley; site 12 consists of detailed sampling across three separate quarry faces (Tuoro quarry) in the same large phonolitic flow that had also been sampled at sites 10 and 11.

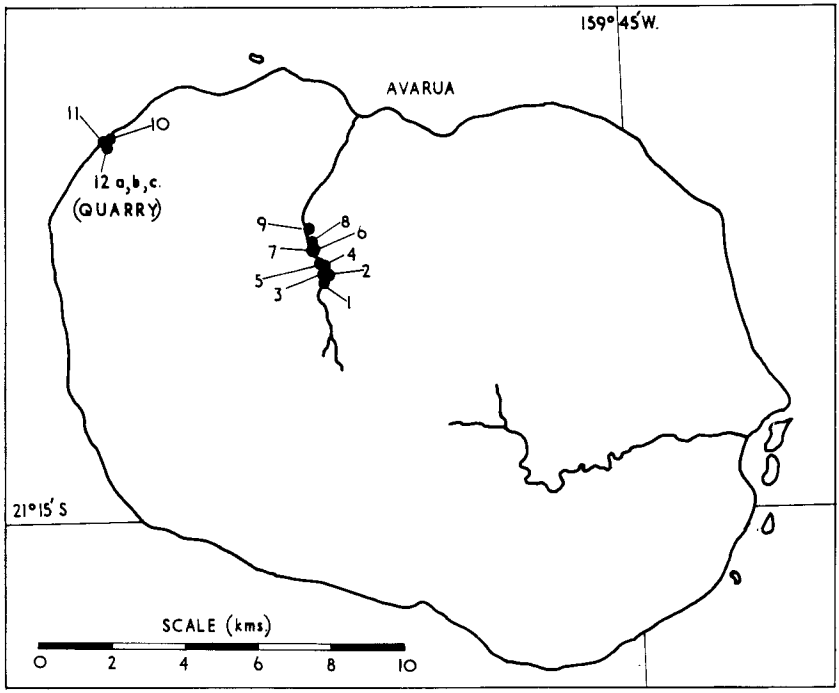


FIG. 1—Location of sampling sites in Rarotonga.

METHODS AND TECHNIQUES

Hand samples were collected in the field and one or more cores were drilled out of each hand sample in the laboratory. Each core was sliced to yield one cylinder, referred to as a specimen. The intensity and direction of natural remanence (NRM) of each specimen were measured with an astatic magnetometer (Blackett, 1952; Collinson *et al.*, 1957), which was also used to determine the susceptibility. Four representative pilot specimens were chosen at random from the collection and these were subjected to partial demagnetisation in incremented alternating magnetic fields (Irving *et al.*, 1961a) and one specimen was also subjected to incremented temperatures (Irving *et al.*, 1961b); the intensity and direction of remanence was re-measured after each increment in treatment. All specimens were measured after treatment in 150 (peak) oersteds in order to reduce the effects of low coercivity components of natural remanence. Two specimens were given a thermoremanent magnetisation by heating above their Curie point and allowing to cool in the Earth's magnetic field, and two specimens were given an isothermal remanent magnetisation.

RESULTS

The initial directions of NRM were scattered but were of positive inclination and southerly declination (Table 1 and Fig. 2). Reduction of low coercivity components decreased the overall scatter and the declination remained southerly and inclination positive.

The four pilot specimens subjected to a.c. demagnetisation showed good stability (Fig. 3), as the stability index* (Tarling and Symons, 1967), given

*Extremely stable >15; very stable >5; stable >2·5; poor stability; >1; metastable >0·5; unstable <0·5.

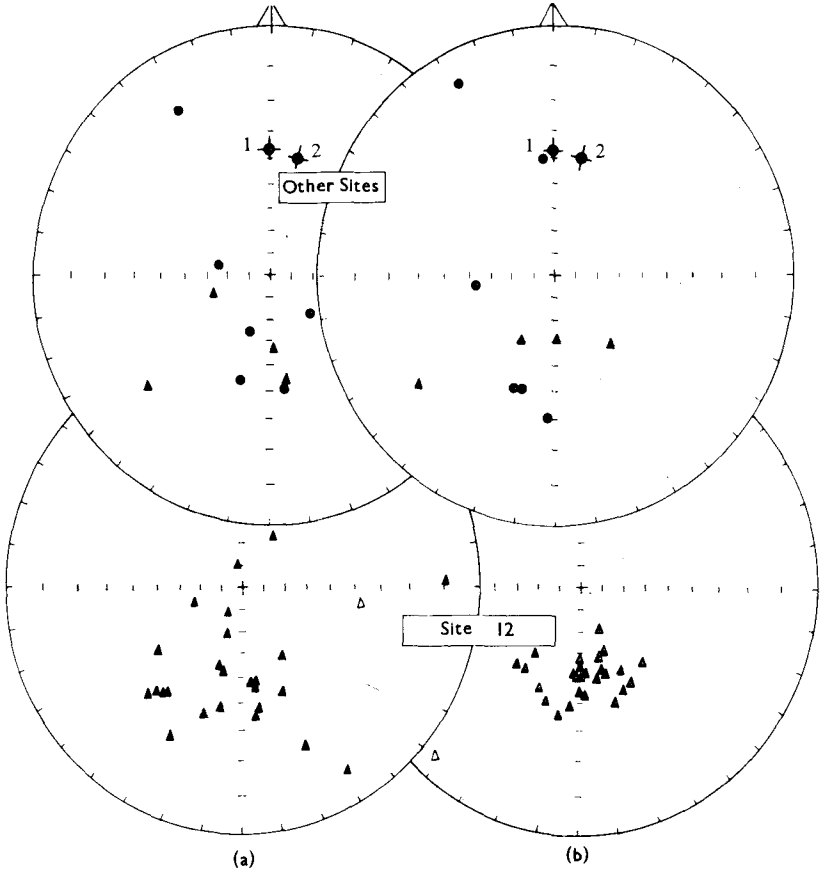


FIG. 2—Stereographic projections of the NRM directions of samples (*a*) before and (*b*) after partial demagnetism in 150 (peak) oersteds. Solid symbols are of positive inclination, hollow symbols are of negative inclination. The projection is equal-angle (Wulff). △ Intrusive; ○ extrusive; 1 is the axial dipole field, and 2 is the Earth's present magnetic field direction in Rarotonga.

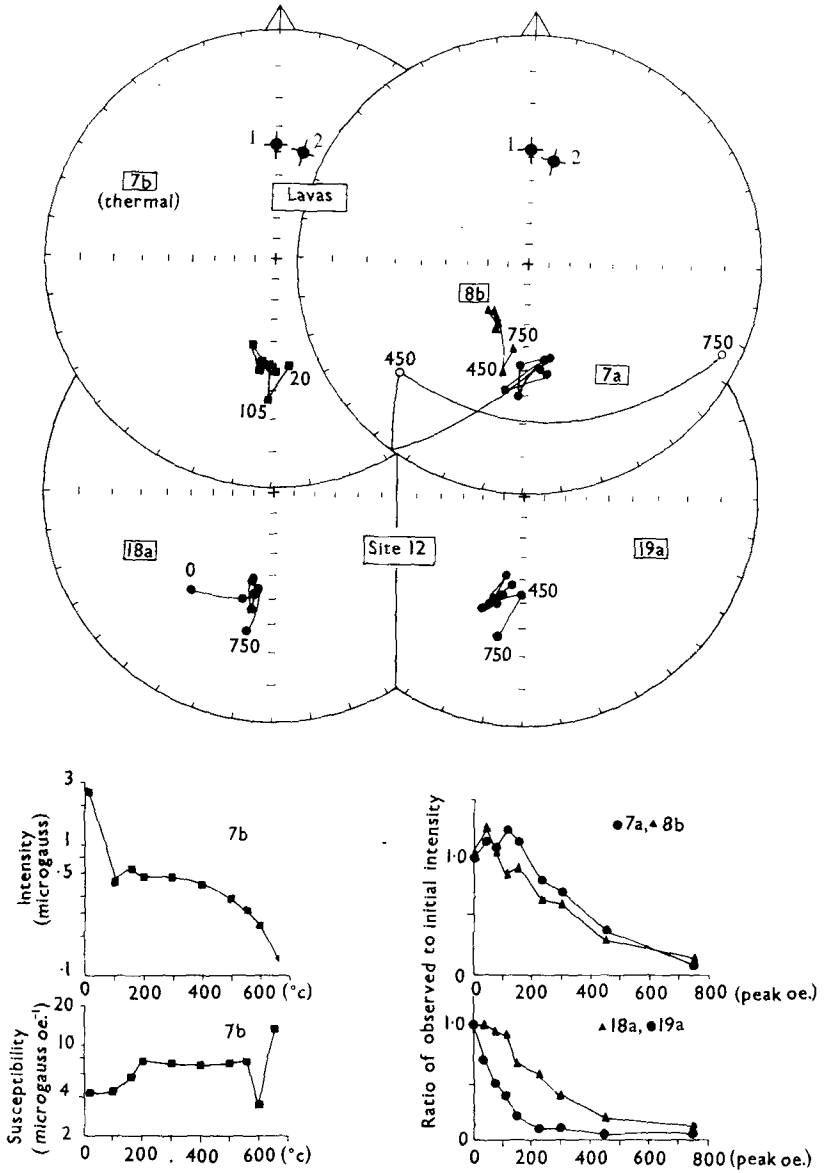


FIG. 3—Partial demagnetisation of pilot specimens. △ Intrusive; ○ extrusive; 1 is the axial dipole field, and 2 is the Earth's present magnetic field direction in Rarotonga.

by max. (range$^{\frac{1}{2}}$/circular standard deviation) for three or more consecutive observations of the remanent vector, corresponded to very stable (5·2) in specimens 8b and 18a, stable (3·4) for specimen 19a, and poor to moderate stability (1·9) for specimen 7a. The behaviour of these pilot specimens also suggested that treatment of the remaining specimens in 150 (peak) oersteds would effectively reduce the effect of any viscous remanence. The least stable specimen (7) was also subjected to partial thermal demagnetisation (Fig. 3) and showed stability (stability index = 4·4) suggesting that partial thermal demagnetisation may have been a more effective technique in isolating the direction of original remanence, although the difference between the results obtained from either technique would be small.

TABLE 1—Mean Site Data

	(*a*) Initial Site					(*b*) Treated (150 oersted)			
	M	X	D	I	Sig.*	D	I	Sig.*	csd.†
1	1·42	5·85	315	+42	25	286	+37	60	72
2	2·01	0·23	229	+21	—	233	+18	—	—
3	6·45	5·12	170	+44	—	140	+50	—	—
4	2·19	11·14	200	+63	—	354	+40	—	—
5	2·37	7·07	132	+65	—	263	+54	—	—
6	2·66	6·07	172	+40	—	183	+30	—	—
7	2·42	2·18	252	+61	—	209	+57	—	—
8	5·29	2·92	176	+57	—	178	+61	—	—
9	5·03	0·69	196	+42	—	197	+39	—	—
10	0·04	—	122	+17	31	151	+49	2	13
11	0·03	—	204	+77	—	179	+50	—	—
12A	0·02	0·03	193	+82	0	169	+56	0	12
12B	0·03	0·03	188	+41	0	180	+48	0	16
12C	0·03	—	213	+30	0	204	+27	3	31

M is the geometric mean intensity of remanence (microgauss) of the specimens. Overall geometric mean intensity of sites 10–12 is 0·028 microgauss and of sites 1–9 is 2·74 microgauss.

X is the susceptibility (microgauss oersted^{-1}) of one specimen per sample in sites 1–9, two specimens in site 12A and one specimen in site 12B. The overall geometric mean susceptibility is 1·43 microgauss oersted^{-1}.

D, I are the mean declination and inclination of the remanent vector, calculated giving equal weight in each sample, (*a*) before and (*b*) after partial demagnetisation in 150 (peak) oersteds.

*Sig. is the significance of the resultant vector (Watson, 1956; Vincenz and Bruckshaw, 1960).

†csd is the circular standard deviation.

Two specimens (8a and 18b) were given a thermoremanent magnetisation (TRM) in 0·59 oersteds. The directions of TRM were both the same as the applied field, suggesting the absence of significant anisotropy, but the intensity of TRM was considerably higher than that of their natural remanent magnetisation (NRM); the ratio NRM/TRM, after the TRM has been corrected to that acquired in 0·42 oersteds (the present intensity of the Earth's magnetic field in Rarotonga) was 0·18 and 0·08 respectively. The low NRM values are thought to reflect a decay of the original TRM rather than to imply a much weaker Earth's magnetic field some 2 m.y. ago.

Specimens 17a and 30a had low maximum intensity of isothermal (IRM) remanence (3·0 and 0·8 microgauss) and moderate coercivity of maximum remanence (225 and 270 oersteds).

The intensity and susceptibility (Table 1) in sites 1–9 were comparable with other Late Cenozoic basaltic lavas (Tarling, 1966); but in sites 10–12, the large dike, both the intensity and susceptibility values reflect a low magnetic mineral content.

CONCLUSIONS

The lavas and dikes in Rarotonga appear to be all reversely magnetised with respect to the Earth's present magnetic field. The mean directions, after parallel demagnetisation in 150 (peak) oersteds, for sites 10–12 are 180°, +48°, with circular standard deviation 20°, and for sites 1–9, 223°, +59°, with circular standard deviation of 47°. Therefore the lavas and dikes sampled in Rarotonga are reversely magnetised. This observation correlates both in time and polarity with similar observations in Hawaii (McDougall and Tarling, 1963) and the Sierra Nevada (Doell *et al.*, 1966) and is therefore consistent with a reversal of the main geomagnetic field, although both dated samples show exsolved ilmenite and titanomagnetite, which may imply self-reversal (Ade-Hall, 1964).

The scatter of sample directions, both within and between sites, is quite large. This could be attributed to secular variation of the geomagnetic field by some 2 m.y. ago in excess of its present magnitude in the Pacific (Gaibar-Puertas, 1951); but in view of the scatter observed within the single large flow (sites 10–12), which is assumed to have cooled over a period of time much shorter than the full secular variation cycle, then the majority of scatter within this site must be attributable to collection and measurement error. This suggests that the overall between-site scatter may be attributed to the effect of this within-site scatter rather than to secular variations. However, more detailed sampling would be of considerable interest in studies of the secular variation of the geomagnetic field in the Pacific.

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